

GR00T N1 · arXiv 2503.14734 · Oct 7 club

Archaeologist

11+3 · seat empty

SigLIP-2+SmolLM2 → Eagle-2 → N1
→ N1.6 → N1.7 → BEHAVIOR'26 || $\pi_{0.5}$

Spine: Eagle-2 System 2 (~10 Hz) + FM DiT System 1 (120 Hz) via cross-attn

H=16 · K=4 · N1 ≠ N1.7 · NOT a classical planner

NVIDIA · Fourier GR-1 · open humanoid VLA

Club one-liner

GR00T N1 = open dual-system humanoid VLA

Eagle-2 System 2 (~10 Hz) + FM DiT System 1 (120 Hz) via cross-attn

$H=16 \cdot K=4 \cdot \text{data pyramid} \cdot \text{path to N1.7} \parallel \pi_{0.5}$ on BEHAVIOR'26

N1 $\neq$ N1.7 · NOT a classical planner

All seats empty this session — still ship full five packs for club/GitHub

Influential priors · CLEAR citations

Archaeologist · speak FOILS + ANCESTRY, not every row

PRIOR	ROLE
SigLIP-2 + SmolLM2	Vision + LLM → Eagle-2
Eagle-2 (Li'25)	System 2 backbone
Lipman FM / π_0 Beta	Action CFM schedule (cite+contrast)
Diffusion Policy / ACT	Generative chunks · IL baselines
DiT / Flamingo / VIMA	DiT blocks + cross-attn pattern
LAPA / IDM / DexMimicGen	Video actions · sim pyramid
π_0 MoE (FOIL)	Shared-attn expert — N1 contrasts
OXE / Octo projectors	Cross-emb data + MLP I/O

Pivot · what N1 locks

1

**Humanoid-first
open FM-VLA**

2

**Dual-system
cross-attn DiT**

3

**Data pyramid
+ neural I2V**

4

**Per-emb MLPs
multi-robot I/O**

Before → After

AR/low-Hz VLAs & single-emb IL → open 2.2B humanoid VLA with 10/120 Hz dual-system packaging
and a reproducible Isaac → N1.7 challenge path

Genealogy MUST · CLEAR one-liner



PIVOT this paper locks

Humanoid-first FM-VLA · explicit dual-system cross-attn DiT · scaled data pyramid

Subsequent: Isaac GR00T · N1.6-BEHAVIOR1k · N1.7 GA · BEHAVIOR'26 dual baseline

TENTATIVE: exact N1.5 internals without primary docs

Lineage · N1 → Isaac → N1.6 → N1.7 → BEHAVIOR'26

CLEAR: N1.7 GA = official co-baseline with $\pi_{0.5}$ · do NOT mix APIs/horizons



VERSION DISCIPLINE

Paper N1 $\neq$ N1.7 (backbone / horizon / package)
N1.6 BEHAVIOR1k numbers $\neq$ N1.7 forecasts

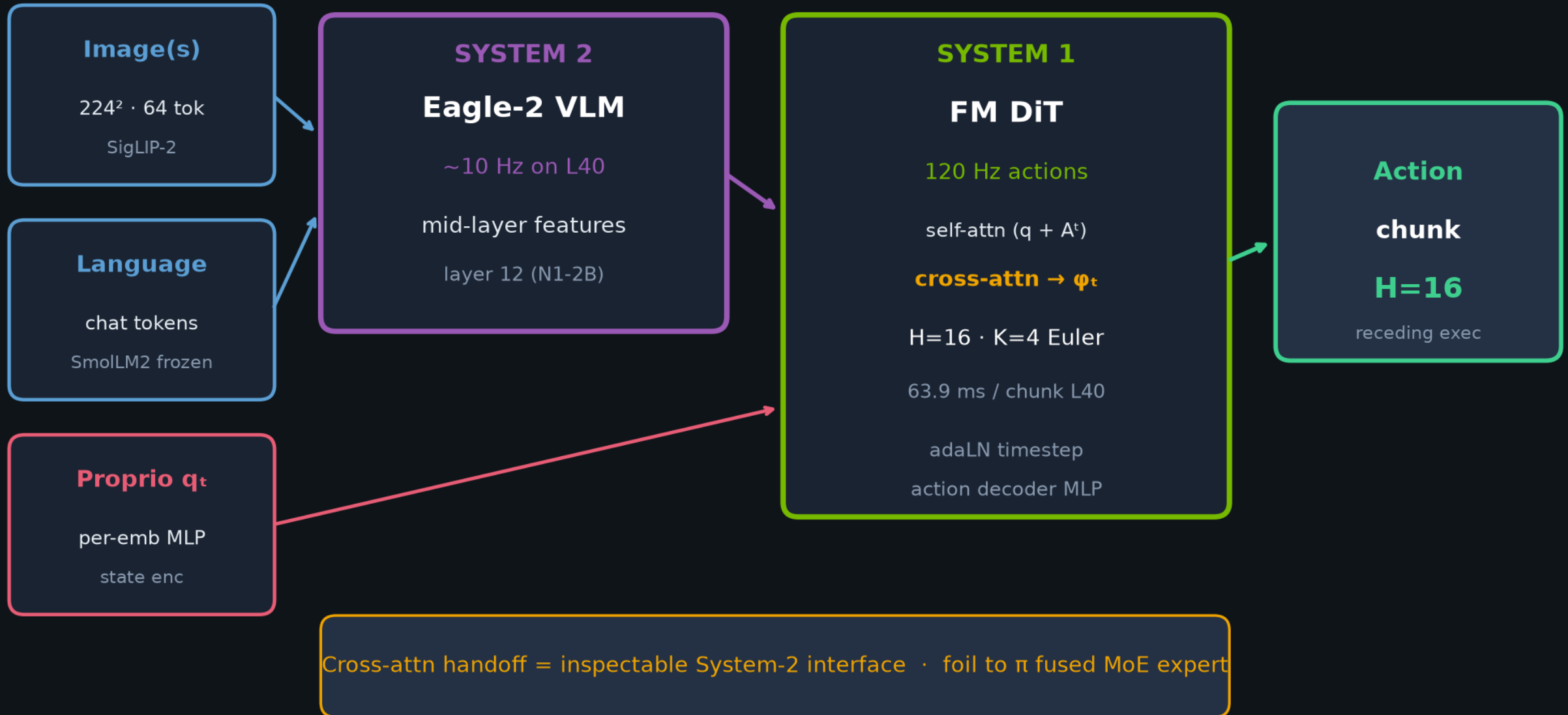
π track (parallel)

$\pi_0 \rightarrow$ OpenPI $\rightarrow \pi_{0.5} +$ RTC
BEHAVIOR'25 forks $\rightarrow$ '26 co-baseline

Genealogy: SigLIP-2+SmolLM2 $\rightarrow$ Eagle-2 $\rightarrow$ N1 $\rightarrow$ N1.6 $\rightarrow$ N1.7 $\rightarrow$ BEHAVIOR'26 || $\pi_{0.5}$

Dual-system VLA · Eagle-2 System 2 × FM DiT System 1

Jointly trained end-to-end · cross-attn bridge · NOT a classical planner



GR00T N1 vs π_0 / $\pi_{0.5}$ · same CFM family, different handoff

BEHAVIOR'26 ships BOTH as official baselines — dual-track is first-class

AXIS	GR00T N1	π_0 / $\pi_{0.5}$
Coupling	Separate DiT · cross-attn	Action expert · shared self-attn
VLM	Eagle-2 (SmolLM2+SigLIP-2)	PaliGemma lineage
Chunk / denoise	H=16 · K=4 · 10/120 Hz	H=50 (π_0) · ~10 Euler · 20-50 Hz
Embodiment I/O	Per-emb MLPs · LAPA/IDM	Pad/mask shared dim
Data bet	Pyramid + neural I2V	OXE-scale + $\pi_{0.5}$ co-train
Successor	N1.7 ~3B · H≤40 · Apache-2.0	$\pi_{0.5}$ + RTC + BEHAVIOR forks

FLAG: modular dual-system / cross-attn DiT vs fused shared-attn MoE expert · same chunk+flow interface

Data pyramid · quantity ↓ · embodiment-specificity ↑

~593M frames / ~8376 h · human + neural + DexMimicGen + real

PEAK · Real robot

GR-1 teleop ~88 h · OXE · AgiBot-Alpha · RH20T

MIDDLE · Synthetic

DexMimicGen ~780k traj · neural I2V 88h→827h · LAPA/IDM

BASE · Web + human egocentric

Ego4D · Ego-Exo4D · EPIC · HOI4D · Assembly-101 · LAPA latent actions

Visionary durable idea: open humanoid data pyramid survives backbone swaps (Eagle → Cosmos-Reason2 → ...)

Control spine · CFM cousin, dual-system packaging

NOT classical planner · generative continuous chunk controller

$$A_t^\tau = \tau A_t + (1 - \tau)\varepsilon \quad \cdot \quad \mathcal{L}_{fm} = \mathbb{E} \|V_\theta - (\varepsilon - A_t)\|^2$$

VL @ 10 Hz

Eagle-2 mid features
cache across denoise

Denoise K=4

Euler on DiT
63.9 ms / H=16

Commit + replan

Don't wait full H
open-loop; recede

Contrast: OpenPI action-expert FM · DP CNN/Transformer · π_{RL} PPO · Comet RFT

Same CFM math as π_0 — different reasoning↔motor handoff

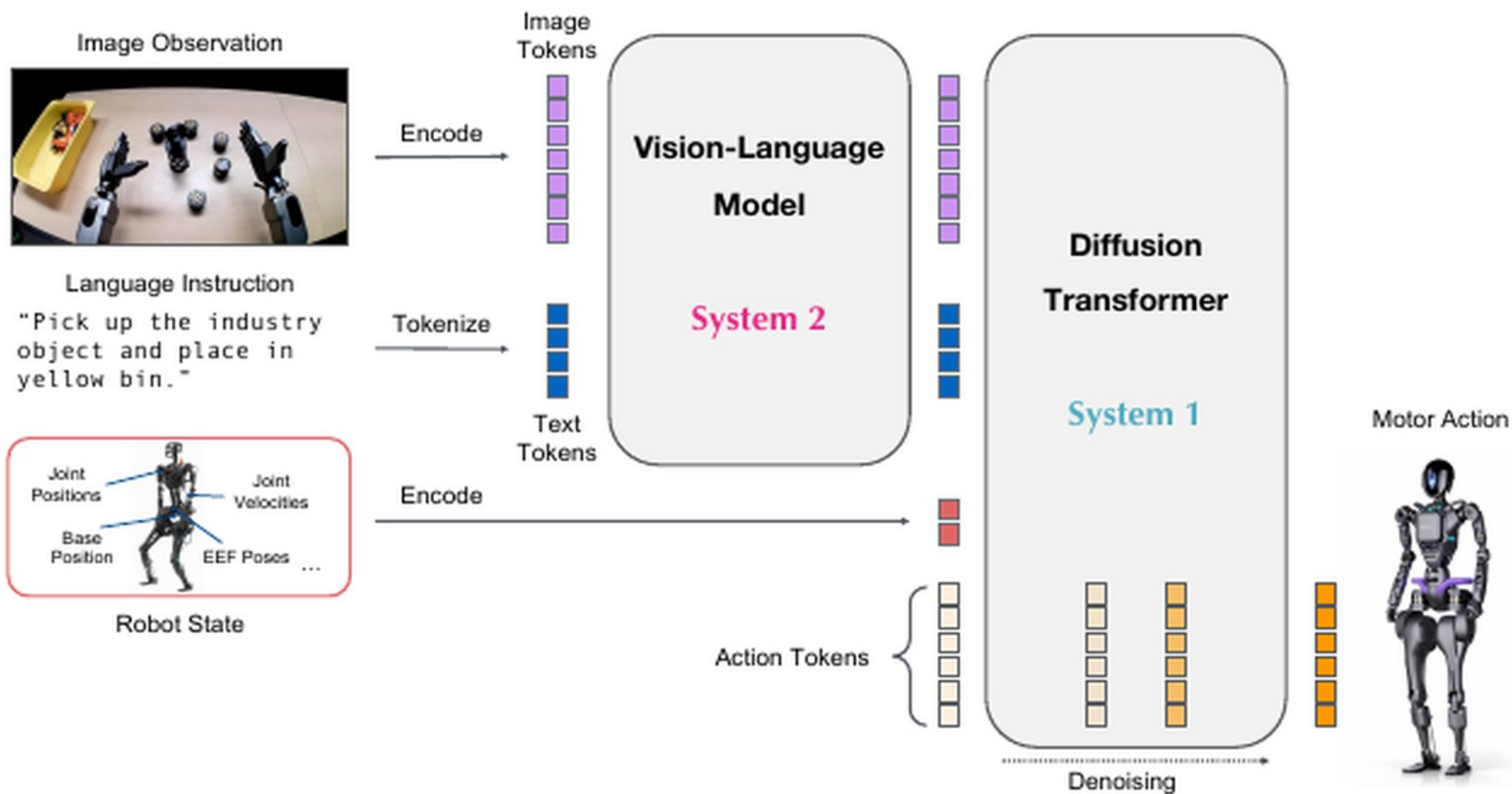


Figure 2: GR00T N1 Model Overview. Our model is a Vision Language Action (VLA) model that adopts a

Archaeologist close

Genealogy one-liner for the room

SigLIP-2+SmolLM2 → Eagle-2 → N1 → N1.6 → N1.7
→ BEHAVIOR'26 || $\pi_{0.5}$

GR00T N1 · Oct 7 · all seats empty · still full five packs

VERSION DISCIPLINE · N1 ≠ N1.7

APIs / horizons / backbones differ — pin the checkpoint you serve

Paper N1

GR00T-N1-2B · ~2.2B (1.34B VLM)

Eagle-2 · SigLIP-2 + SmolLM2

H=16 · K=4 · mid-layer 12

63.9 ms / chunk @ L40 bf16

Table 2/3 tattoo numbers live [HERE](#)

N1.7 GA

nvidia/GR00T-N1.7-3B · ~3B

Cosmos-Reason2-2B (Qwen3-VL)

H ≤ 40 · new package / API

Apache-2.0 · BEHAVIOR'26 baseline

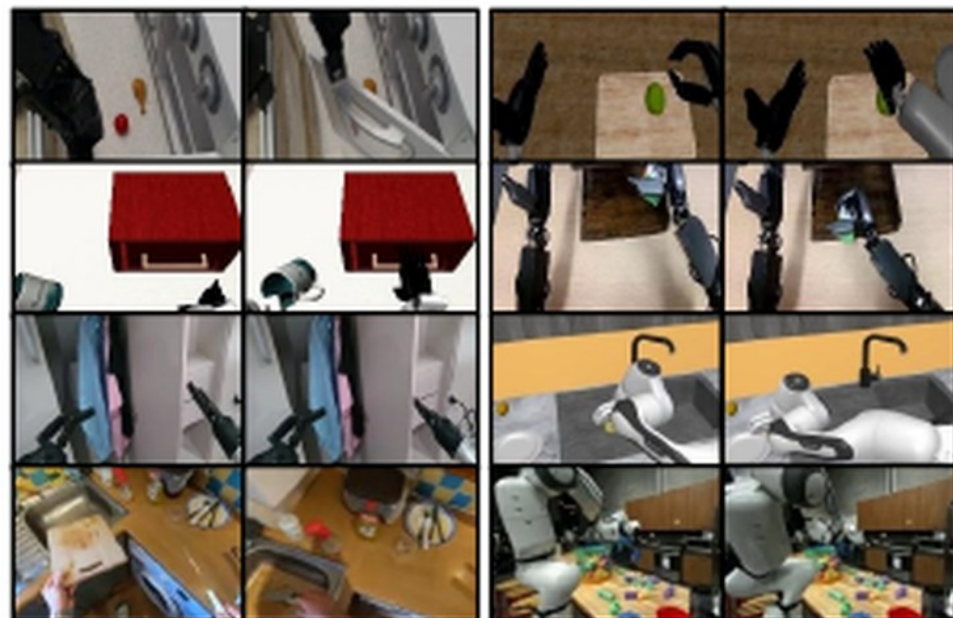
Do NOT quote N1 tables as N1.7

Stakeholder / Visionary rule

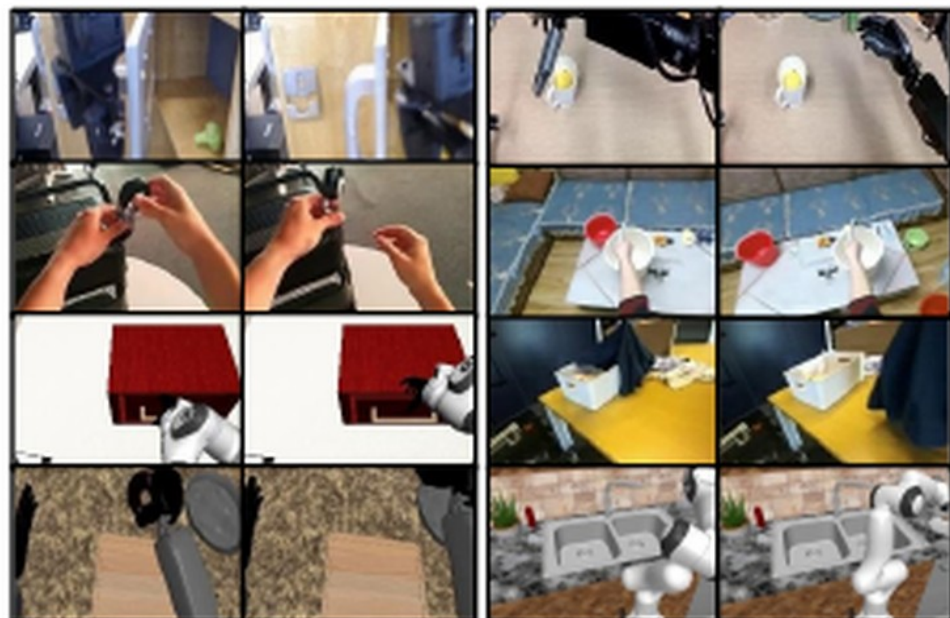
Prefer finetune N1.7 for product/challenge · use N1-2B only for paper replication

Match served horizon to trained horizon · N1.6 BEHAVIOR1k ≠ N1.7 forecast

Also: N1 is NOT a classical planner — generative CFM chunk controller with dual-system packaging



Latent action 1: Move the right arm to the left



Latent action 2: Move the right arm to the right

Figure 4: **Latent Actions.** We retrieve similar latent embeddings across various embodiments. The left images illustrate the latent action that corresponds to moving the right arm (or hand) to the left, while the right images illustrate the latent action that corresponds to moving the right arm (or hand) to the right. Note that this general latent action is not only consistent in different robot embodiments, but also in human embodiment.